

Project ID :

25-26J-402

1. Topic (12 words max)

AI-Powered Intelligent Education Support System

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-402
Year	2025-2026

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The traditional education system continues to face challenges in meeting the diverse needs of modern learners. Rigid curricula, one-size-fits-all teaching strategies, and limited feedback mechanisms hinder the ability to address individual differences in learning pace, background, and accessibility. As classrooms grow more diverse, there is a critical need for intelligent, data-driven educational systems that can adapt to each student's unique learning profile. Although existing e-learning platforms provide access to digital content, they often lack real-time adaptability, personalized learning paths, and proactive intervention capabilities [1].

This research proposes the development of an **AI-Powered Intelligent Education Support System**, designed to integrate multiple advanced components to improve learning outcomes, monitor student progress, and assist educators in decision-making. The system will include modules such as an adaptive learning path generator, emotion and engagement detection via computer vision, academic risk prediction through behavioral analytics, and a smart feedback engine. These modules aim to deliver personalized learning experiences, provide early identification of at-risk students, and support targeted interventions to enhance academic success [2], [3].

Furthermore, the system will incorporate assistive technologies such as **visual learning tools for differently-abled students**, using computer vision and text-to-speech functionalities. For example, a webcam-based object recognition module can convert real-world items into labeled visuals and spoken words, assisting students with learning disabilities in building vocabulary and recognition skills [4].

By applying cutting-edge AI/ML techniques—including reinforcement learning, natural language processing (NLP), facial emotion recognition, and sentiment analysis—the proposed system will not only improve academic performance but also support emotional well-being and inclusiveness [5]. It addresses the growing need for continuous, real-time educational support in both traditional and remote learning environments.

The novelty of this research lies in the **holistic integration of intelligent modules** within a single, unified platform. Unlike standalone tools, this system functions as a dynamic educational ecosystem—driven by real-time data, learner behavior, and performance metrics. It adapts continuously to individual needs, supports diverse learners, and enhances teaching effectiveness. Ultimately, the project aims to contribute to the advancement of intelligent tutoring systems and inclusive education technologies, paving the way for more personalized, equitable, and efficient learning environments.

References

- [1] L. Chen, P. Chen, and Z. Lin, “Artificial intelligence in education: A review,” *IEEE Access*, vol. 8, pp. 75264–75278, 2020, doi: 10.1109/ACCESS.2020.2988510.
- [2] R. S. Baker and P. S. Inventado, “Educational data mining and learning analytics,” in *Learning Analytics*, Springer, 2014, pp. 61–75, doi: 10.1007/978-1-4614-3305-7_5.
- [3] B. P. Woolf, *Building intelligent interactive tutors: Student-centered strategies for revolutionizing e-learning*, Morgan Kaufmann, 2010.
- [4] V. Kumar and C. Rose, “Intelligent tutoring systems for inclusive education,” in *International Handbook of Inclusive Education*, Springer, 2019, pp. 439–455.
- [5] A. I. Alzahrani and M. Habib, “Emotion detection using deep learning for educational improvement,” *Education and Information Technologies*, vol. 27, no. 4, pp. 4723–4743, 2022, doi: 10.1007/s10639-021-10703-1.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

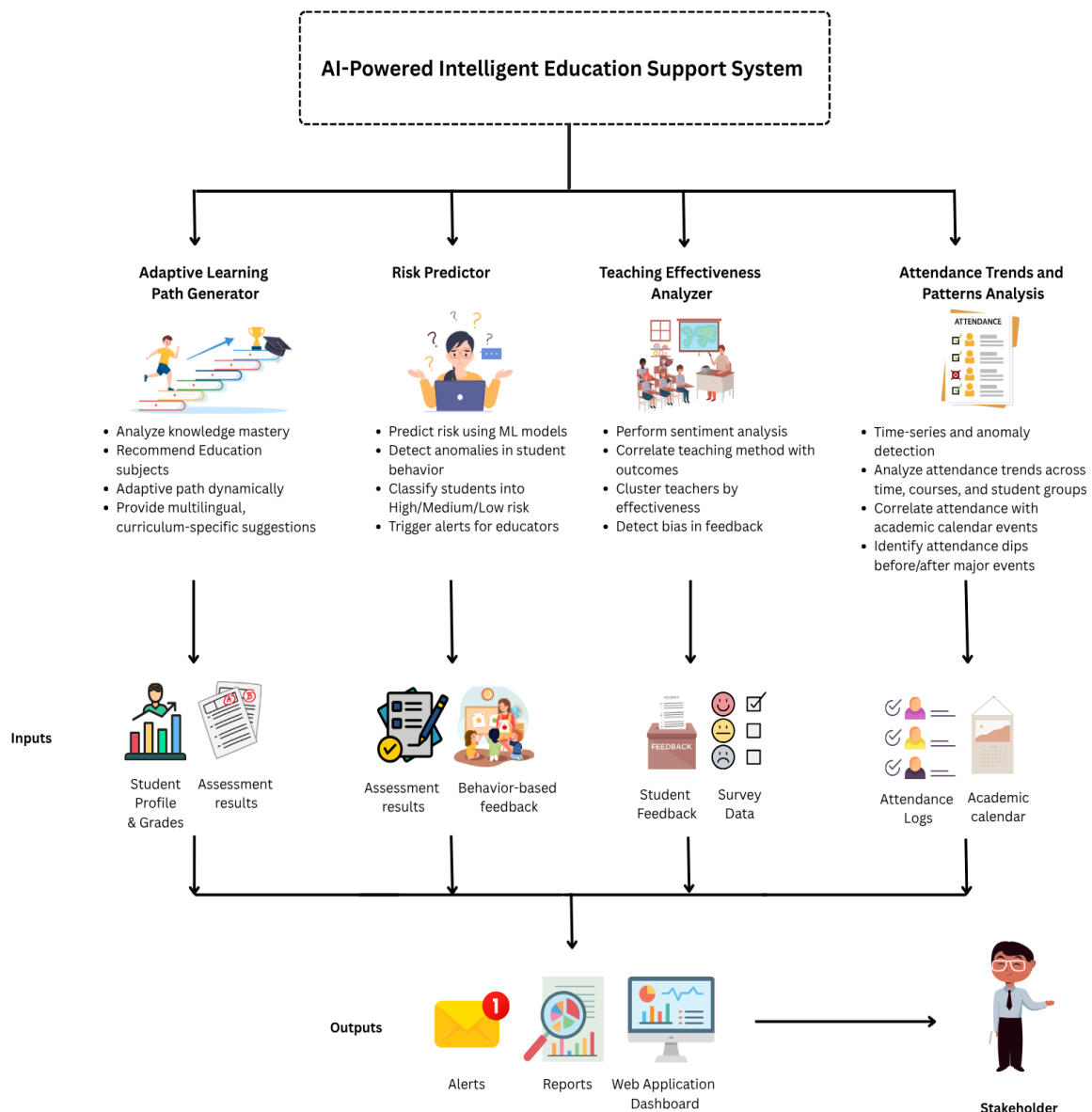
The proposed solution is an AI-powered intelligent education support system designed to improve student learning, identify academic risks, evaluate teaching quality, and understand attendance patterns. It works as a web-based platform that uses student data from different sources like online learning systems, quizzes, attendance records, and feedback to provide smart, personalized support to both students and educators.

One of the main parts of the system is the Adaptive Learning Path Generator. This component creates a unique learning path for each student based on their current knowledge and skills. As students complete lessons and assessments, the system tracks their progress and recommends the next topics or exercises that match their level. This helps students learn more effectively and at their own pace.

The second part is the Academic Risk Early Warning System. This system looks at behavior and performance data—such as attendance, grades, and participation—to detect students who may be struggling. When it finds a risk, it sends alerts to teachers so they can give support in time.

The Attendance Trend Analyzer evaluates attendance records using time-series forecasting and anomaly detection techniques to identify patterns, critical risk periods, and correlations with academic performance.

All modules are connected through a centralized data processing engine that integrates input from learning platforms, feedback systems, and academic databases. The system is designed to be scalable, multilingual, and adaptable to various curricula, offering a robust framework for intelligent, inclusive, and personalized education.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

To successfully develop and implement the AI-powered Intelligent Education Support System, expertise is required in several specialized domains. First, a strong understanding of **educational theory and pedagogy** is essential. This knowledge helps in designing personalized learning paths, setting appropriate content difficulty levels, and aligning recommendations with curriculum standards. Educational psychologists and instructional designers can contribute valuable insights into student learning behaviors and motivation factors.

Second, **artificial intelligence and machine learning expertise** is crucial. Developers need to apply techniques such as knowledge tracing, reinforcement learning, sentiment analysis, classification, time-series analysis, and clustering. These methods help in modeling student knowledge, predicting academic risks, analyzing teacher performance, and detecting attendance trends. Experience in training and validating machine learning models using real-world educational data is important to ensure accuracy and reliability.

Natural Language Processing (NLP) skills are also needed, especially for analyzing qualitative feedback from students about their learning experience and teacher performance. In addition, **data visualization** expertise is useful for presenting insights through dashboards and reports in a way that educators can easily understand and act upon.

Regarding data requirements, the system needs access to a variety of student-related data. This includes academic performance data (e.g., quiz and exam scores), behavioral data (e.g., attendance records, LMS activity logs), and feedback data (e.g., open-ended comments or survey responses). It is also important to ensure data privacy and security, especially when handling personally identifiable information. Ethical guidelines and institutional approvals must be followed when collecting and processing educational data.

Overall, this project demands cross-domain collaboration between education experts, AI/ML engineers, software developers, and data analysts to ensure the system is both technically effective and educationally meaningful.

8. Objectives and Novelty

Main Objective To develop an AI-powered Intelligent Education Support System that enhances personalized learning, accurately predicts academic risks, evaluates teaching effectiveness, and analyzes attendance patterns, thereby improving overall student outcomes and supporting data-driven decision-making in educational institutions			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22907820 R P A Prabuddhini	Capable of analyzing student attendance data across courses and time frames to identify trends, anomalies, and correlations with academic performance.	Attendance Data Aggregation Collect and organize attendance records by day, course, semester, and other relevant time frames. Trend and Anomaly Detection Use time-series forecasting models (ARIMA, LSTM) and anomaly detection algorithms to identify unusual attendance patterns. Correlation Analysis Analyze relationships between attendance dips and external events or academic calendar factors. Visualization	Predictive Attendance Modeling Combines forecasting and anomaly detection for early identification of attendance risks and behavioral changes. Contextual Correlation Analysis Links attendance trends with academic events and external factors to provide deeper insights beyond raw attendance numbers.

		Present attendance data through heatmaps, charts, and dashboards for actionable insights.	Actionable Visualizations Offers intuitive visual tools that assist administrators in planning targeted interventions and resource allocation.
IT22590626 B P D M K Ariyathilaka	Capable of analyzing students' academic records for key subjects and providing evidence-based predictions on their performance in the upcoming G.C.E. (O/L) exam. It focuses on identifying at-risk subjects, evaluating factors affecting results, and recommending methods to improve pass rates.	Data Collection and Integration Gather attendance records, term test marks records for all subjects Risk Classification Use supervised machine learning models (e.g., Logistic Regression, Random Forest, SVM) to classify students into risk categories based on the aggregated data. Behavioral Trend Analysis Apply time-series analysis to monitor changes in student behavior over time, refining risk predictions. Alert Generation Develop a notification system that delivers prioritized alerts to educators for timely interventions.	Multi-Source Dynamic Risk Prediction Integrates heterogeneous academic and behavioral data sources for a holistic and continuously updated risk assessment. Severity-Based Alerts Prioritizes notifications based on risk severity to prevent alert fatigue and ensure critical cases receive immediate attention. Explainable Predictions Uses explainable machine learning

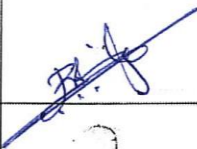
			techniques to provide interpretable risk factors, aiding educators in understanding and acting on alerts.
IT22565730 K M S Dulshan	Capable of dynamically personalizing students' learning journeys by continuously assessing their knowledge levels and skill mastery to recommend the most relevant topics and learning materials. And this system helps students find their academic learning journey according to their performance.	Data Aggregation Collects term test marks in a few years, from students aged 13–17 using school systems and digital platforms to support personalized learning. Personalized Content Recommendation Develop models that analyze ongoing assessment data to recommend customized learning materials and exercises tailored to individual student needs. Mastery Tracking Continuously track student mastery of individual topics and adjust future lessons accordingly to optimize learning progression. Multi-language and Curriculum Support	Real-Time Adaptive Learning Combines deep knowledge tracing and reinforcement learning to deliver continuously updated, personalized learning paths, surpassing static or one-size-fits-all approaches. Cross-Curriculum and Multilingual Flexibility Supports diverse educational standards and languages, enabling use across different regions and educational systems. Explainable Recommendations

		Implement support for various languages and curriculum standards to ensure accessibility and scalability. Real-Time Feedback Provide immediate feedback to students and educators on learning progress and suggested next steps.	Utilizes explainable AI to provide transparency behind content recommendations, increasing trust and engagement.
IT22571984 H P R D T Wickramarathna	To predict the stress level of students before major exams by analyzing their study habits, sleep patterns, workload, and recent performance, and to provide personalized recommendations for stress management.	Collect data on study hours, sleep hours, pending assignments, and recent marks (via forms or LMS logs). Preprocess and engineer features (study–sleep ratio, workload index, marks trend). Train a classification model (Random Forest / SVM) to categorize students into Low, Medium, High stress levels. Evaluate model performance and visualize stress trends through dashboards. Generate personalized recommendations (e.g., reduce study load, improve sleep, revise weaker subjects).	This component focuses on the well-being aspect of students rather than purely academic outcomes. Unlike traditional performance prediction, it integrates behavioral and lifestyle data (study hours, sleep, workload) with academic results to forecast stress. The novelty lies in providing early warnings and actionable advice to reduce exam burnout — an area less addressed student support systems.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22907820 R P A Prabuddhini	Attendance Trend Analyzer The component uses time-series forecasting (ARIMA, LSTM) and anomaly detection, which are advanced data analysis tasks rooted in IT. The ability to mine insights from data and visualize trends using IT tools like dashboards and visual analytics is essential in fields like data science, business intelligence, and academic software systems. The creation of data pipelines, visualization components, and predictive models reflects the IT skillset in software engineering, data analytics, and AI applications.
IT22590626 B P D M K Ariyathilaka	Academic Risk Prediction System This involves data integration, supervised learning (ML), classification algorithms (SVM, Random Forest), and dashboard development, all of which are key technical capabilities under the IT domain. The use of predictive analytics reflects real-world IT applications in educational data mining and student behavior modeling. The system's design and implementation involve programming, statistical analysis, and deployment of models, aligning perfectly with IT specialization skills.
IT22565730 K M S Dulshan	Adaptive Learning Path Generator This component demonstrates strong application of Artificial Intelligence (AI), particularly Reinforcement Learning, Knowledge Tracing, and Data Analytics, which are core areas in modern IT. It requires developing intelligent algorithms that process real-time assessment data to personalize educational content directly reflecting the use of IT in intelligent systems development, personalized software, and machine learning applications.
IT22571984 H P R D T Wickramarathna	Exam Stress Level Prediction This component applies AI/ML techniques to predict a student's stress level before major exams using behavioral and academic factors such as study hours, sleep hours, pending assignments, and recent marks. By classifying students into Low, Medium, or High stress categories, the system provides early alerts and personalized recommendations to manage workload and reduce burnout. It highlights the use of classification algorithms (Random Forest, SVM), feature engineering, and dashboard visualization, reflecting real-world applications of AI in student well-being monitoring.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Lokesha	Weerasinghe	
Co-Supervisor	Prof.	Dilshan	De Silva	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.